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SUPERSEDING
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30 July 1985

MILITARY SPECIFICATION

CRANE, MULTI-PURPOSE, RAILWAY, ON/OFF TRACK, 12,000 POUNDS CAPACITY

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a multi-purpose railway crane with car spotter capability designed for operation on rails, on paved surfaces, or on unimproved rough-terrain surfaces with a 360 degree swing cab, with a lifting capacity of 12,000 pounds and can be equipped with optional attachments for boom and vehicle.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-V-173	- Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-3558	- Plate, Identification, Locomotive, Railway Cars, and Work Equipment.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-162	- Materials Handling Equipment: Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limits for Army Materiel.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems.

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Vehicle Safety Standards and Regulations.
Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the

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issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Manual of Standards and Recommended Practices, Mechanical Div.

Section C, Part II, Volume I - Specification for Design, Fabrication and Construction of Freight Cars, M-1001.
Section C, Part II, Volume II - Appendices M-1001. Plate B - Clearance Diagram.

(Application for copies should be addressed to the Association of American Railroads, American Railroads Bldg., 1920 L Street, N.W., Washington, DC 20036.)

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

B30.15 - Mobile Hydraulic Cranes.
B93.28 - Method for Calibration of Liquid Automatic Particle Counters Using "AC" Fine Test Dust.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel.
D14.1 - Specification for Welding Industrial and Mill Cranes.

(Application for copies should be addressed to the American Welding Society, 2501, NW 7th Street, Miami, FL 33125.)

FEDERAL RAILROAD ADMINISTRATION (FRA)

(Application for copies should be addressed to the Federal Railroad Administration, 400 7th Street, SW, Washington, DC 20024.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC., (SAE)

SAE Handbook.

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(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The crane is a multi-purpose railroad maintenance vehicle with a boom and cab capable of rotating 360 degrees and a maximum rated lifting capacity of 12,000 pounds at 10-foot radius. The crane shall be in compliance with safety standard ANSI B30.15 for mobile hydraulic cranes and shall be both a rail type, and a rough terrain type vehicle. The crane shall be equipped with four single flanged railwheels for operation on 56-1/2-inch gage track, and four pneumatic-tired wheels for rough terrain operation on unimproved and paved road surfaces. A minimum of one axle shall oscillate at least 8 degrees in either direction from a horizontal axis. The crane shall be capable of operating the attachments provided by the contractor. In addition to the attachments specified in 3.12, the following attachments shall be made available upon order by individual Government activities that are users of this end item.

Boom Attachments

13-20 ft adjustable boom extension
1-1/4 cu yd bucket
9 Ft wide track cleaning bucket
8 in cribbing bucket
Forks
Brush cutter
3 and 4 cu yd snow buckets
18" and 42" clam shell
12' depth backhoe
Fork tie bailer
Wheel set grapple

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of the crane from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish cranes conforming to this specification.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or

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of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice.

3.3.1 Material deterioration prevention and control. The cranes shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the crane may be newly fabricated from recovered materials to the maximum extent practicable, provided the crane produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the crane.

3.4 Maintainability. The crane shall be designed so that accessories and components are readily accessible for repair or replacement with minimum removal or disturbance of adjacent parts or components, using general-purpose tools. Operating parts, accessories, and drain outlets shall be readily accessible for regular maintenance service. Drainage of all liquids shall be by gravity into ground level containers.

3.5 Identical items. All cranes furnished shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviations will be acceptable without written approval of the contracting officer.

3.6 Performance. The crane shall be capable of the following:

a. Crossover exposed rails, and transferring from road to rail and rail to road operation, where the running surface of the trackage is 9 inches above the adjacent surface.
b. Moving over 500 feet three or more railway cars having a minimum distributed gross weight of 300 tons while operating on rail.
c. Maintaining a constant speed of not less than 8 miles per hour (mph) for not less than 1/2 mile while towing a single 50-foot railway car having a minimum gross weight of 125 tons.
d. Without connected load, attaining a speed not less than 17 mph, in either direction of travel on rails, and a speed not less than 20 mph forward on level pavement.

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e. By means of the towing clevis, handling a trailer with a starting tractive effort on average paved surfaces of not less than 3000 pounds.
f. Lifting a load of not less than 12,000 pounds, at 10-foot radius from centerline of rotation, vertically raising the 12,000-pound load at least 13 feet in not more than 10 seconds.
g. Swinging the boom with cab through 360 degrees between a time of 20 seconds minimum to 30 seconds maximum while engine is at governed speed and swing control is at fully opened position.
h. Swinging the boom through 180 degrees from side to side with a load of 7400 pounds at 10-foot radius while on the rail on the single-flanged wheels.
i. Being able to rail and derail on a grade of 10 percent or more without loss of stability or control. The grade need not incorporate rail.
j. Functionally using the rail and tie attachments with no sign of damage or deformation to crane or attachments.

The rated load capacity shall not be more than 75 percent of the tipping load for any operating position, and there shall be no sign of damage or permanent distortion after operation at maximum capacity. Maximum application of the lowering control of the rail guide wheels of the crane shall not cause slippage of the driving wheels at maximum torque, or derailing at switches and crossings.

3.7 Design and construction. The design of the crane shall be in accordance with the best engineering practice for this type of equipment. All moving or high temperature parts that are of such nature or so located as to become a hazard to operating or maintenance personnel, shall be fully enclosed or properly guarded. Exhaust shall be directed away from the cab. All parts requiring service, or subject to replacement, shall be easily accessible. The crane shall be rigidly constructed to provide support for all operating subassemblies in proper alinement. All parts shall be shaped and assembled to function as a crane. Provisions shall be made to determine from the operator's seat when the rail-drive wheels are accurately located for lowering onto the rails. Handholds, platforms, and railings shall be provided, in conformance with SAE J185, where needed for the safety of operating personnel engaged in getting on or off, and during approach and coupling operations. All controls and instruments shall be arranged for convenient operation and observation about the operator's seat and to provide for one-man control of all operations on road or rails. Maximum use shall be made of ball or roller bearings. When bronze bushings are used, means shall be provided to prevent turning out of alinement with lubricant channels. Loads shall not be applied at the caps or joints of split half bearings.

3.7.1 Clearances. When self-propelled or towed on track, the vehicle shall be designed, and any components attached to the vehicle shall be designed, to lock in a position so as to not interfere with track, signal, crossing or other structures on or along a railway. The maximum width for road travel shall not exceed 104 inches. Projections under the chassis shall have ample clearance to prevent interference, when mounting or crossing the rails (see 3.6).

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3.8 Couplers. A standard fully operating AAR coupler shall be provided at one end of the vehicle and shall be adjustable in height to allow coupling while on rail, or on road, or both. Air brake connection hose assembly shall be installed; and means shall be provided inside for operating this coupler. Another coupling device shall be installed at the other end of vehicle and, while vehicle is on rail, this coupling device shall be designed to mate with railway car couplers conforming to requirements of the AAR.

3.9 Towing pintle. A towing pintle, suitable for attaching towing tongues with commercial clevis type and Military Lunette ring attachment devices, shall be centrally located for road service on the rear of the crane. The pintle shall not be loosened or damaged by a test pull of 4000 pounds applied through a swing from 45 degrees left to 45 degrees right from the line of crane travel.

3.10 Air brake system. An AAR railway car air braking system, complete with air hose connections shall be provided, so that one or more moving cars can be brought to a stop without assistance from the crane braking system. The air compressor shall be provided with an air intake filter, and shall be powered by an air-cooled gasoline engine with electric starting motor, or by a power takeoff from the crane engine or by means of a hydraulic motor. The air braking system shall be complete with receiver (air reservoir), safety valve, control piping, and drain. An automatic brake valve, with air pressure gage, for rail car brake control, shall be located in the cab. The air piping shall be arranged to drain condensate into the reservoir, and when such arrangement is impracticable, a suitable centrifugal dirt and moisture collector shall be applied to all points where dirt and moisture may collect.

3.11 Service and parking brakes. The crane shall be equipped with foot actuated hydraulic service brakes, and either foot operated or hand controlled parking brakes, for both rail and road use. On rails, the stopping distance from 17 mph to a dead standstill shall not exceed 100 feet, and on road, the stopping distance from 20 mph to a dead standstill shall not exceed 72 feet. The road service brakes shall conform to the DOT Motor Carrier Safety Regulation, sections 393.40 through 393.42, and 393.45 through 393.48 or SAE J1152. The parking brake shall hold on either road or rails against a horizontal line pull not less than 4000 pounds.

3.12 Attachments. The following attachments shall be furnished with each crane:

A tiltable boom extension capable of handling a 39-foot section of rail over the front of the machine by picking up the rail at its midpoint.
Tie grapple, hydraulically operated.
Rail tongs, hydraulically operated.
Boom extension, 10-foot minimum length for tie grapple and rail tongs.

3.13 Lights and signals. The crane shall be provided with a minimum of two sealed beam automotive type headlights, and backup lights, suitably located to permit night railing and derailing operations with or without the aid of auxiliary lights. The road driving lights and reflectors shall conform to DOT Motor Carrier Safety Regulations, sections 393.12, 303.19, 393.20, 393.22

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through 392.26(d), 393.27 through 393.31, and 393.33. The turn signal operating unit shall have visible and audible flash indicators. For rail operation, a headlight shall be located facing boom so operator can see without additional lighting for any position of boom. The instrument panel shall be adequately illuminated and one light with a switch shall be installed inside the cab.

3.14 All-weather cab. The crane shall have an enclosed operating cab provided with safety glass windows. The door(s) shall have keylock(s) and weather seals. The windshield shall be provided with automatic wipers. The cab shall be heated by means of a heater and defroster with air circulation fan(s). Heating facilities shall have adequate capacity and direction control to defrost and de-ice not less than 75 percent of the cab window areas in ambient temperatures of -10° Farhenheit (F), and shall maintain the temperature in the cab at not less than 50 °F. Metal frames, upholstered seats with upholstered back rests shall be furnished. The upholstery shall be of a serviceable, weather-resistant material. The seat shall be located and provided with mirrors, for unobstructed view and as required for the safe performance of all intended functions. The cab shall meet the requirements of SAE J1040C for rollover protective structures for wheeled industrial front end loaders.

3.15 Engine. The crane shall be powered by an industrial two- or four-cycle diesel engine. The engine shall be of a commercial design that has proven to be satisfactory in extensive use, and shall be one for which repair parts are readily obtainable. The intermittent brake horsepower and speed rating shall be in excess of that required to meet the operational requirements of the crane. The engine shall be complete with the accessories normally furnished, including not less than the following:

a. A 12 volt (V) electrical system including either an electric or hydraulic starting system, either battery charging generator of not less than 30-ampere (A) rating or an alternator of not less than 35-A rating, voltage regulator, and voltmeter.
b. An engine housing.
c. A fuel tank of sufficient capacity for 8 hours of normal operation.
d. A fuel filter.
e. A speed governing system with provision for adjusting the speed setting.
f. An engine hour meter.
g. A fuel level gage.
h. A lubricating oil pressure gage.
i. A cooling liquid temperature gage.
j. A volt meter.
k. Torque converter temperature gage.
l. An air cleaner for engine combustion air inlet.
m. An engine block heater, a hydraulic oil tank heater, and an engine oil pan heater shall be installed.

3.15.1 Transmission and torque converter. The transmission shall be provided with three selective hydraulically controlled driving ranges in each direction for rail travel, and not less than three forward speeds and reverse for road travel. The drive chains or shafts, differentials, and axle shafts shall be

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matched to and be compatible with the engine, and shall have a torque capacity exceeding the maximum delivered engine torque developed through the intervening gear reductions and torque multiplications.

3.15.2 Noise limits.

3.15.2.1 Personnel noise. The noise produced by the crane shall conform with MIL-STD-1474 requirements when tested in accordance with 4.5.16.1. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3 and be readable at the distance indicated on the sign.

3.15.2.2 Exterior noise. The crane shall meet the requirements of MIL-STD-1474 for exterior noise for construction and materials handling equipment when tested in accordance with 4.5.16.2.

3.16 Radio interference suppression. The unit shall be suppressed for radio interference in accordance with SAE J551. Suppressors used shall conform to SAE J552.

3.17 Fungus resistance. Electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.18 Hydraulic system.

3.18.1 Components. A hydraulic system with a fixed displacement pump, directly driven by or from the engine, shall be provided. The system shall be complete with all necessary parts including pump, relief valve, fluid reservoir, gages, control valves, and piping for the operation of road and rail-guide wheel raising and lowering rams, railway car coupling positioner(s), steering control booster, boom and any other hydraulically operated equipment. The capacity of the system shall be adequate for continuous operation of all connected equipment at top efficiency. Hydraulic hoses, tubing, components, and parts shall be of material and size to permit the free flow of hydraulic fluid with a line velocity limit not to exceed 22 feet per second. Provisions shall be made for bleeding the system at all necessary points. Tubing shall conform to SAE J524 or J525, Low Carbon Steel Tubing for Bending and Flaring. Where flexing is required, hydraulic hose shall conform to SAE J517, size, type, and pressure rating as required. The hose shall be protected against damage at all points of wear. Reusable straight thread swivel fittings shall be used. Full-flow strainers shall be furnished and located between the hydraulic tank suction intake and the pump. The strainers shall have a reusable corrosion resistant element that will prevent passage of all particles of a size larger than 150

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microns. The system shall withstand a maximum pressure of not less than 1-1/2 times the normal operating pressure required for operation of the crane, without leaks, deformation, or breakage. A return line filter shall be provided.

3.18.2 Contamination levels (hydraulic cleanliness). The particulate matter in hydraulic fluid removed from the system directly upstream of the filter, shall not exceed the ISO code level of 19/16 in accordance with SAE J1165 after completion of assembly of the crane when tested as specified in 4.5.3.

3.19 Wheels and tires. The road wheels shall be a commercial, heavy-duty tractor type, selected for the impact loading of crossing exposed railroad tracks. Tires shall have individual rate load-carrying capacities equal to the maximum individual tire loading imposed by the operation. The tires shall be a low pressure type to provide for safe operation over rails at ungraded crossings. Normal operating air pressure shall not exceed the tire or rim manufacturer's ratings.

3.20 Toolbox. Means shall be provided to carry such tools and accessories as are required to operate the crane. Toolboxes shall be made of steel, not less than 0.0751-inch (US revised standard gage No. 14) thick. The toolbox shall have a hinged lid and a truck drawbolt or hasp of a type that will keep the lid closed when the toolbox is subjected to vibration, and shall be provided with padlock and steel chain. The toolbox shall be securely fastened to the crane in a protected and accessible location.

3.21 Nameplate. A nameplate conforming to MIL-P-3558 shall be furnished by the contractor for each crane. The contractor shall securely affix the nameplate to each unit in a conspicuous place.

3.22 Instruction plates. Each crane shall be equipped with instruction plates, including warnings and cautions, suitably located, describing any special or important procedures to be followed in operating and servicing. Instruction plates shall be of a copperbase alloy, painted steel or aluminum alloy. Instruction plates shall meet the durability requirements in SAE J115A for permanent signs.

3.23 Lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a qualified products list (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.23.1 Lubrication. Bearings and moving parts requiring lubrication shall have means provided for such lubrication. Pressure lubrication fittings shall not be employed where normal lubricating pressure may damage grease seals or other parts unless pressure relief is provided. The crane shall be lubricated prior to delivery, and tagged in a conspicuous place to identify the lubricants and their temperature range.

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3.24 Treatment and painting. The portions of the crane normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383.

3.25 Technical publications. All technical publications, parts manuals, renewal parts catalogues, instructional materials, maintenance procedures, and instructions normally supplied shall be furnished.

3.26 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with accepted commercial practice for automotive-locomotive equipment.

3.26.1 Steel fabrication. Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

3.26.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.26.3 Riveted connections. Rivet holes shall be accurately punched or drilled, and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of uniform size and shape for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with rivet holes, and in full contact with the surface of the member.

3.26.4 Welding. Welding shall be in accordance with AWS D1.1, AWS D14.1, and AAR Manual of Standards and Recommended Practices, Section C, Part II or other applicable requirements. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermittent welds for strength will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided, when necessary, to reduce stress concentration.

3.26.4.1 Welders. Before assigning any welders or welding operator to weld work, the contractor shall obtain certification that the welder or welding operator has passed qualification tests in accordance with ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualification. Certification shall be available for review (upon request) by the contracting officer or designated representative.

3.27 Endurance. While on rail, the crane shall be capable of completing 50 hours and a minimum of 1350 cycles within that time of the rail test as specified in 4.5.18. While on road, the crane shall be capable of completing 50 hours and a minimum of 900 cycles within that time of the road test as specified

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in 4.5.18. For both rail and road tests, a load of not less than 5000 pounds shall be lifted vertically at 10-foot radius from centerline of rotation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection. Prior to examination and test of the crane specified in 4.3.1 and 4.3.2, the following shall be performed:

a. Service the crane with oils, greases, and fuel specified herein and designated for use in the ambient temperatures at which the tests will be conducted.
b. Conduct the run-in as prescribed (see 4.5.2.2).
c. Furnish a schedule of maintenance to be followed during all testing of the cranes. A list of tools required to perform this maintenance shall also be provided.

4.3.1 First article examination.

4.3.1.1 Pretest examination. Prior to testing, all first article cranes shall be examined for the defects marked "x" in column 1 of table I. Presence of one or more defects shall be cause for rejection.

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TABLE I. Examination schedule.

First article	Quality conformance	Defects	Requirement paragraph
X		101. Design and construction not as specified.	3.7
X		102. Clearances not as specified.	3.7.1
X	X	103. Lights and signals not as specified.	3.13
X	X	104. Materials not as specified.	3.3
X	X	105. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.	3.3.1
X	X	106. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.3.2
X	X	107. Contractor does not have documentation available for identification of material, material finishes or treatments.	3.3.3
X	X	108. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the crane.	3.3.4
X		109. Maintainability not as specified.	3.4
X	X	110. Identical items not as specified.	3.5
X	X	111. Service and parking brakes not as specified.	3.11
X	X	112. Attachments unavailable or not furnished as specified.	3.1, 3.12
X	X	113. All-weather cab not as specified.	3.14
X	X	114. Engine not as specified.	3.15
X	X	115. Transmission and torque converter not as specified.	3.15.1
X	X	116. Hydraulic system not as specified.	3.18.1
X	X	117. Wheels and tires not as specified.	3.19
X	X	118. Instruction plates not as specified.	3.22
X	X	119. Lubricants not as specified.	3.23
X	X	120. Lubrication not as specified.	3.23.1
X	X	121. Treatment and painting not as specified.	3.24
X	X	122. Technical publications not furnished as specified.	3.25
X	X	123. Welding not as specified.	3.26.4
X	X	124. Couplers not as specified.	3.8
X		125. Radio interference suppression not as specified.	3.16
X	X	126. Fungus resistance not as specified.	3.17
X	X	127. Toolbox not as specified.	3.20

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TABLE I. Examination schedule. (Cont'd)

First article	Quality conformance	Defects	Requirement paragraph
X	X	128. Nameplate not as specified.	3.21
X	X	129. Workmanship not as specified.	3.26
X	X	130. Steel fabrication not as specified.	3.26.1
X	X	131. Bolted connections not as specified.	3.26.2
X	X	132. Riveted connections not as specified.	3.26.3
X	X	133. Paint color not as specified.	3.24
X	X	134. Welder and welding operator certification not as specified.	3.26.4.1

4.3.2 First article tests. Upon successful completion of the examination specified in 4.3.1.1, one first article crane shall be subjected to the tests marked "x" in column 1 of table II. Except as otherwise specified herein, test conditions shall apply throughout testing. Failure of any test shall be cause for rejection.

TABLE II. Test schedule.

First article	Quality conformance	Tests	Test paragraph	Requirement paragraph
1	2	3	4	5
X		Test conditions.	4.5.2.1	
X	X	Run-in.	4.5.2.2	
X	X	Hydraulic cleanliness.	4.5.3	3.18.2
X		Load test.	4.5.5	3.6
X		Air brake test.	4.5.6	3.10
X		Speed test.	4.5.7	3.6
X		Rail brake test.	4.5.8	3.11
X	X	Road brake test.	4.5.9	3.11
X		Hi-rail test.	4.5.10	3.6
X	X	Parking brake.	4.5.11	3.11
X	X	Towing pintle.	4.5.12	3.9
X	X	Railing test.	4.5.13	3.6
X	X	Boom swing.	4.5.14	3.6
X	X	Boom lift.	4.5.15	3.6
X		Noise.	4.5.16	3.15.2
X		Heater.	4.5.17	3.14
X		Endurance test.	4.5.18	3.27

4.4 Quality conformance inspection.

4.4.1 Examination. Each crane shall be examined for the defects "x" in column 2 of table I (see 4.5.1). Presence of one or more defects shall be cause for rejection.

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4.4.2 Tests. Each crane shall be tested as specified in column 2 of table II (see 4.5.2). Failure of any test shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. Cranes shall be examined as specified herein in accordance with table I. An "x" in the column indicates the examinations that shall be conducted.

4.5.2 Tests. Tests shall be conducted in accordance with table II. An "x" in the column indicates the tests that shall be conducted.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at the ambient temperature and climatic conditions existing at the place of test. Only that maintenance established by the contractor and submitted as maintenance schedule prior to commencement of the tests shall be performed during the tests. Safety and human factors considerations shall be evaluated throughout testing. Operation of the crane shall be with fuels, oils, lubricants, and antifreeze as applicable for the temperature conditions existing at place of test.

4.5.2.2 Run-in. Before starting the tests, the engine shall be warmed up to operating temperature. The proper air pressure for braking, and the proper hydraulic system fluid pressure shall be indicated on the gage. Tire pressure and all lubrication points shall be checked, all operating controls shall be checked for proper functioning. Each crane shall then be driven through a run-in for not less than 30 minutes. The run-in shall include operating in all gears, operating all functional equipment, and performing all hydraulic functions. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any component, or evidence of leakage of air, fuel, lubricant or hydraulic oil, shall be cause for rejection of the crane by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after the deficiency is corrected.

4.5.3 Hydraulic system roll-off cleanliness. Test each crane in accordance with test method No. 1. Continue cleanup procedure until the average of three consecutive counts conform to 3.18.2.

4.5.4 Rail test site. The rail test site shall be provided by the contractor, unless otherwise specified by the Government, and shall include straight level track, a yard spur switch, a finish grade crossing, and a level approach with the ungraded surface approximately 9 inches below the running surface of the rails.

4.5.5 Load test. The crane shall couple with 3 cars with a minimum gross weight of 300 tons and be capable of moving them on straight level track from rest for a distance not less than 500 feet. The crane shall also move a 125 ton load for 1/2 mile at a speed of not less than 8 miles per hour (mph).

4.5.6 Air brake test. The crane with 3 moving cars with a minimum gross weight of 300 tons shall be stopped without assistance from the crane braking system.

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4.5.7 Speed test. The crane, without a connected load, shall attain a speed of not less than 17 mph in forward and reverse on straight and level track, and a speed of not less than 20 mph forward on level pavement.

4.5.8 Rail brake test. On straight and level track the crane, without a connected load, shall stop within 100 feet from a speed not less than 17 mph in forward and reverse.

4.5.9 Road brake test. On level pavement the crane shall stop within 72 feet from a speed not less than 20 mph in forward.

4.5.10 Hi rail test. The crane shall perform not less than two crossings, transfer from road to rail and transfer from rail to road at a point where the running surface of the rails is between 8 and 10 inches above the adjacent surface.

4.5.11 Parking brake. The parking brake shall be tested by means of a measured pull of not less than 4000 pounds in both directions of travel on rails and in both directions on level pavement.

4.5.12 Towing pintle. The towing pintle shall be tested by means of a measured pull of not less than 4000 pounds applied through a swing from 45 degrees left to 45 degrees right from the line of crane travel.

4.5.13 Railing test. The crane shall demonstrate the railing and derailing procedure on a grade of at least 10 percent without loss of stability or control. This grade need not incorporate rail.

4.5.14 Boom swing test. The crane boom, while the engine is at governed speed and swing control is at the fully opened position, shall swing 360 degrees between a time of 20 seconds and 30 seconds.

4.5.15 Boom lift test. The crane, while on rails, over the front, shall pick up and move a load of 12,000 pounds at a 10-foot radius from centerline of rotation and shall pick up a load of 7,400 pounds at a 10-foot radius and swing 180 degrees from side to side at least two times. A test weight of 12,000 pounds and 7,400 pounds shall be made available by the contractor.

4.5.16 Noise measurement.

4.5.16.1 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load, MIL-STD-1474, 5.1.2.1.4; contours shall be taken at not fewer than 12 equal horizontal arc increments; one increment shall include data from the noisiest position; additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.16.2 Exterior noise test. Measure the exterior noise level in accordance with SAE J88. Nonconformance to 3.15.2.2 shall constitute failure of this test.

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4.5.17 Heater. The heater shall be tested in accordance with SAE J1129. Nonconformance to 3.14 shall constitute failure of this test.

4.5.18 Endurance test. The crane shall be tested in accordance with test method No. 2, Railtest, and test method No. 3, Road Test, for the length of time, cycles and loads as specified in 3.27. The crane shall perform all the operations as specified, without failure, damage, or permanent deformation and shall perform all tests without exceeding temperature limits for the cooling system, engine oil, transmission oil, and hydraulic system. Nonconformance to 3.27 shall constitute failure of this test.

4.6 Inspection comparison. The Government may select cranes at any time during the contract-production period and subject these cranes to the examination specified in 4.6.1 and column 1 of table I, and to the tests marked "x" in column 1 of table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government and will not include the previously inspected first article crane. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison crane to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison crane(s) and shall be cause for refusal by the Government to continue acceptance of production cranes until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on cranes yet to be produced by the contract. Any deficiencies found as a result of the inspection comparison will be considered prime facie evidence that all cranes accepted prior to the completion of inspection comparison are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer. Where cranes previously accepted by the Government are determined to be deficient, the contractor will negotiate with the Government to determine the method of correction.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection. Each packaged crane shall be examined for the applicable defects as specified in MIL-STD-162, appendix B. In addition, the following defects shall apply for level C:

135. Exposed, unpainted critical surfaces not preserved.
136. Fragile components not protected as specified.
137. Removed components not secured to crane without increase in cube as specified.
138. Fluid reservoirs not filled with recommended fluids as specified.

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139. Cooling system not preserved as specified.
140. Cranes not in a mobile, operable condition as specified.

Any defects shall be corrected prior to offering the crane for shipment.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each complete crane, including accessories, shall be preserved in accordance with the applicable requirements of MIL-STD-162, for level A, type I mobile shipment. The engine cooling system shall be preserved using the anti-freeze and water procedure of MIL-E-10062.

5.1.2 Level C. The cranes shall be preserved in a manner which will prevent damage and deterioration during delivery to the first destination or to the user. Cranes shall remain operable with their own power. Any exposed, unpainted surfaces considered critical shall be protected with a preservation coating. Fragile components or those subject to pilferage shall be protected with cushioning or blocking, or by removal and placement in containers respectively. Any fluid reservoirs shall be filled to the operating level with the manufacturer's specified lubricants. The engine cooling system shall be preserved as specified for level A.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Each preserved crane shall be packed in accordance with MIL-STD-162.

5.2.2 Level C. Each preserved crane shall be in a mobile, operable configuration. Any removed equipment, whether boxed or not, shall be secured to the crane so as not to increase the cubage of the vehicle or otherwise interfere with movement or handling.

5.3 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129 for level A and C.

6. NOTES

6.1 Intended use. The crane is intended as a multipurpose vehicle for use in and around rail yards for a variety of railroad maintenance tasks depending on the attachments acquired and for locating moving railway equipment in areas along the track which are not suitable for truck traffic. The mobility of the crane on rails, unimproved and paved surfaces, make it suitable for moving any type of wheeled equipment where and as desired.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article crane is to be furnished for inspection and approval (see 3.2).

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c. Color required when other than as specified (see 3.24).
d. Levels of preservation and of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the items should be an initial production model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Lubricants. Selection of lubricants, fluids, and greases shall be in accordance with section 5 of MIL-STD-838, Lubrication of Military Equipment. This prescribes the policy for using specification-type products whenever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113 Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. Unless otherwise authorized by the US Army Belvoir Research, Development, and Engineering Center (ATTN: STRBE-VF), lubricants, fluids, and greases for ground equipment shall be restricted to those listed under section 2 of MIL-HDBK-113.

6.5 Subject term (key word) listing.

Crane
On/Off track
Multi-purpose railway crane
Twelve thousand pound capacity crane

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
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Preparing activity:
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APPENDIX

TEST METHOD NO. 1

HYDRAULIC SYSTEM ROLL-OFF CLEANLINESS

1. TEST APPARATUS

Liquid automatic particle counter.

2. TEST PROCEDURE

a. Connect the hydraulic lines at the cylinders to bypass those cylinders whose capacity is less than 1/2 of the connecting line volume.
b. Run the engine at no-load governed speed pumping the oil through each circuit for not less than 5 minutes. Perform an inline particle count as described in d. below.
c. When applicable, connect each line to its related component. Operate each function of the system for 5 minutes continuously through full stroke and at maximum rates. Perform the particle count as described in (d) below.
d. The procedure for determining contamination shall be as follows:

(1) Contamination levels shall be determined by particle counts. The particle counter used for evaluation of the samples shall be calibrated in accordance with ANSI Recommended Standard Method for Calibration of Liquid Automatic Particle Counters using "AC" Fine Test Dust (ANSI B93.28). The required counts shall be the average of not less than three consecutive counts. Samples may be taken from the reservoir or upstream of the filter(s) used for cleanup.
(2) Use any combination of on-the-crane and off-the-crane filtration for cleanup.
(3) Connect the counter sensor to the system, using a bypass line and a branch line. The bypass line shall connect to the system, have a flow rate not less than 5 times the sensor flow rate, and may return fluid directly to the reservoir. The branch line to the sensor shall tee into the bypass line and shall be not more than 12 inches in length. Any flow restriction devices shall be placed downstream of the particle counter sensor. Start engine and operate at not less than 1000 rpm. Record particle counts.

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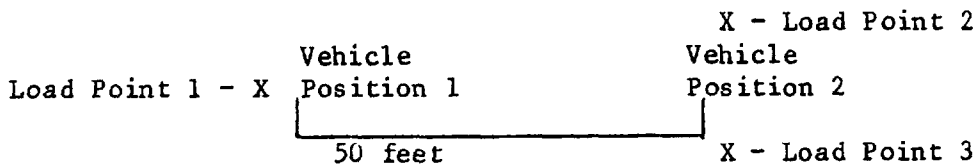
APPENDIX

TEST METHOD NO. 2

RAIL TEST

1. TEST COURSE

The test course shall be on level track as shown in overhead view figure as follows:



2. TEST PROCEDURE

a. The test procedure is as follows: Crane shall start at position 1, pick up load at point 1 in front of crane, move backward on rail 50 feet to position 2, swing load to one side 90 degrees, lower load to ground, pick up load, swing load 180 degrees to the other side of crane, lower load to ground, pick up load, move 50 feet forward to position 1 and lower load to ground at point 1 in front of crane. This completes one cycle.

b. Repeat procedure A for the time limit, cycles, and load as specified.

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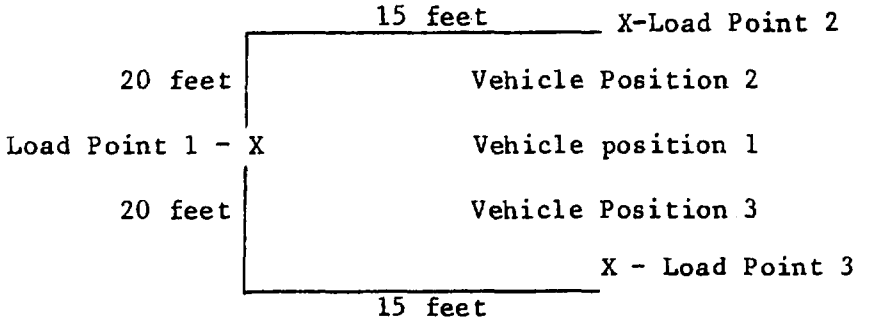
APPENDIX

TEST METHOD NO. 3

ROAD TEST

1. TEST COURSE

The test course shall be on level pavement as shown in overhead view figure as follows. Vehicle position 2 and 3 do not need to parallel vehicle position 1.



2. TEST PROCEDURE

a. The test procedure is as follows: Crane shall start at position 1, pick up load at point 1 in front of crane, move to position 2, swing load to right side 90 degrees, lower load to ground, pick up load, move to position 1, lower load to ground, pick up load at point 1 in front of crane, move to position 3, swing load to left side 90 degrees, lower load to ground, pick up load, move to position 1, lower load to ground at point 1. This completes one cycle.

b. Repeat procedure A for the time limit, cycles, and load as specified.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-53058A(ME)	Crane, Multi-Purpose, Railway, On/Off Track, 12000 Pounds Capacity

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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